

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC)

ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ...

CS XI

Chapter 1: Computer Systems

Lecture 09 of 12

★ Golden

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

LEARN IT

The **Software Development Life Cycle (SDLC)** is a structured process that guides software development from beginning to end. It ensures **clear goals, better planning, risk reduction and high quality**. Without it, teams forget features, go over budget and deliver buggy software.

SDLC has **6 phases**:



Phases 1–3 in detail

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Phases 4–6 in detail

Phase	Activities and details	Deliverable
4. Testing	Find and fix bugs (errors), using Black Box and White Box testing.	Test Reports, bug-free software
5. Deployment	Release the software to end-users: Direct, Phased, or Pilot release.	Live Software
6. Maintenance	Provide support and updates, and fix post-release issues.	Patches, Updated Versions

Deployment methods: In a **direct** release the old system is switched off and the new one switched on at once. In a **phased** release the new system is introduced part by part. In a **pilot** release one small group (for example one branch or one class) uses it first before everyone else.

Testing methods

Black Box Testing

- The tester does **not** see the code.
- Tests functionality: give inputs, check outputs.
- Done by dedicated testers.

White Box Testing

- The tester **can** see the code.
- Tests the internal logic and structure.
- Done by developers.

WORKED EXAMPLE

Building a house = SDLC.

Phase 1 (Requirements): you ask the family what rooms they need — 3 bedrooms, a kitchen, a garage.

Phase 2 (Design): the architect draws the blueprints.

Phase 3 (Implementation): the builders lay the bricks.

Phase 4 (Testing): an inspector checks the wiring and the plumbing.

Phase 5 (Deployment): the family moves in.

Phase 6 (Maintenance): repairs, painting and a new room added later.

WORKED EXAMPLE

Black box vs white box on a login screen. A black-box tester types a correct password and a wrong password and checks that the first is accepted and the second rejected — without looking at the code. A white-box tester (the developer) reads the code and checks that every `if` branch, for example the "account locked after 3 tries" branch, actually runs correctly.

REMEMBER IT

Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass box (you can see inside).

COMMON MISTAKE

Do not say the SDLC ends at deployment. Maintenance is a full phase and often lasts the longest. Also, the SRS is the deliverable of Requirement Analysis, not of Design.

اردو میں سمجھیں

ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ٹیسٹنگ، تنصیب (Deployment) اور دیکھ بھال (Maintenance)۔ پہلا مرحلہ سب سے اہم ہے اور اس سے SRS دستاویز بنتی ہے۔ بلیک باکس ٹیسٹنگ میں کوڈ نظر نہیں آتا جبکہ وائٹ باکس ٹیسٹنگ میں کوڈ دیکھ کر اندرونی منطق جانچی جاتی ہے۔

EXAM TIP

"Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of each (SRS, Design Document, Source Code, Test Reports, Live Software, Patches). A short question often asks you to differentiate black box and white box testing — answer in a 3-row table (sees code?, what is tested?, who does it?).

CHECK YOURSELF

1. Which phase produces the SRS document?
2. Who usually performs white box testing?
3. Name the three ways to deploy software.

Answers: 1. Requirement Analysis. 2. Developers. 3. Direct, Phased and Pilot release.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. SDLC is:

- a. A structured process from idea to retired software
- b. A logic gate
- c. A search algorithm
- d. A chart type

MCQ 2. The first SDLC phase is usually:

- a. Requirement analysis
- b. Maintenance
- c. Coding only
- d. Retirement

MCQ 3. A deliverable is:

- a. The document or product of a phase
- b. A Boolean 1
- c. A pie slice
- d. An IP port

MCQ 4. Testing comes:

- a. After implementation, before deployment
- b. Before requirements
- c. Instead of design
- d. Never

MCQ 5. Maintenance is:

- a. Fixing and updating after release
- b. The first interview
- c. NAND
- d. Big O

MCQ 6. SDLC exists to:

- a. Control quality, cost and time
- b. Replace Boolean algebra
- c. Draw K-maps
- d. Sort arrays

Extra short questions

1. Define SDLC.
2. List the six phases in order.
3. Name one deliverable of requirements and of design.
4. Why not jump straight to coding?

Extra long question

1. Tabulate the six SDLC phases with activities and one deliverable each. ★ Golden 10-mark.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Software Development Life Cycle: a structured process that guides software from beginning to end.
2. Requirements, design, implementation/coding, testing, deployment, maintenance.
3. SRS / requirements document; design diagrams or architecture document.
4. Wrong requirements become expensive bugs; SDLC catches them on paper first.

Long

1. Phase | activities | deliverable. Stress that testing uses the requirements as the oracle, and maintenance is the longest phase in real life.

Next lecture: 1.2.4 – 1.2.7 Waterfall vs Agile Models & Case Studies